Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```
        int j = high;
        int pivot = a[(low + high) / 2];
        int temp;

        while (i <= j) {
            while (a[i] < pivot)
                i = i + 1;

            while (a[j] > pivot)
                j = j - 1;

            if (i <= j) {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;

                i = i + 1;
                j = j - 1;
            }
        }

        if (low < j)
            quickSort(a, low, j);

        if (i < high)
            quickSort(a, i, high);
    }
}
```

```c
int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int i = 0;

    while (i < n) {
        scanf("%d", &a[i]);
        i = i + 1;
    }

    quickSort(a, 0, n - 1);

    i = 0;
    while (i < n) {
        printf("%d ", a[i]);
        i = i + 1;
    }

    return 0;
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.